Assignment–02: Comparative Study and Practical Analysis of Generative AI Models

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1. Introduction

This report compares four generative models, the Autoencoder (AE), Variational Autoencoder (VAE), Deep Convolutional GAN (DCGAN) and a pretrained diffusion model (Stable Diffusion), and analyses outputs from ready-made Google Colab implementations. AE and VAE are studied for image reconstruction (CO3); DCGAN and Stable Diffusion for image generation (CO4).

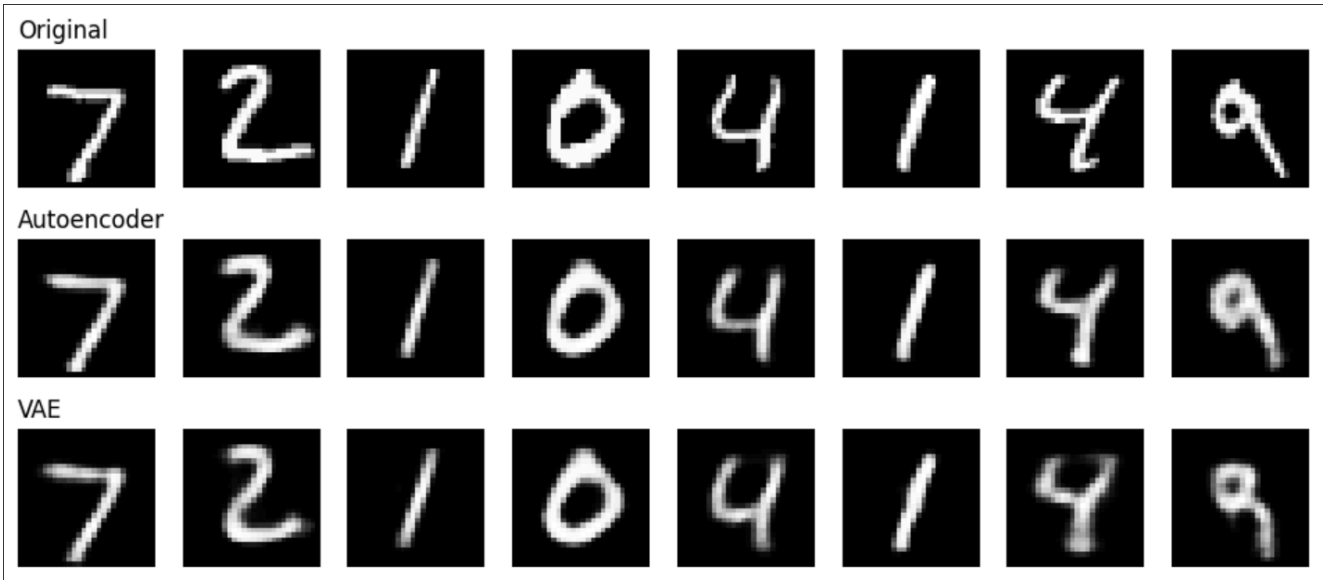
2. Task 1 – Comparative Study

Criteria	Autoencoder (AE)	Variational AE (VAE)	DCGAN	Pretrained Diffusion (Stable Diffusion)
Working principle	Encoder compresses the input into a fixed latent vector; decoder rebuilds it. Trained to minimise reconstruction error.	Encoder outputs mean (μ) and variance (σ^2); z is sampled using the reparameterisation trick and decoded. Loss = reconstruction + KL divergence.	Generator turns random noise into images; discriminator judges real vs fake. Both train against each other (min–max game).	Forward process adds noise step by step; a network learns to reverse it. Stable Diffusion denoises in a compressed latent space guided by a text prompt.
Architecture & major components	Conv/dense encoder → bottleneck (latent code) → mirrored decoder.	Encoder → μ and log σ^2 layers → sampling layer → decoder.	Generator: transposed convs + BatchNorm + ReLU, tanh output. Discriminator: strided convs + LeakyReLU, sigmoid output.	CLIP text encoder, U-Net denoiser with cross-attention, VAE encoder/decoder, noise scheduler.
Advantages	Simple and fast to train; good for compression, denoising and feature learning.	Smooth, continuous latent space; can generate new samples and interpolate.	Sharp, realistic samples; generates in one forward pass.	Highest quality and diversity; stable training; text control; high resolution.
Limitations	Irregular latent space, so poor at generating new images; may memorise data.	Blurry outputs; trade-off between reconstruction and KL term; posterior collapse.	Unstable training, mode collapse; usually low resolution; no text control.	Slow (many steps), heavy GPU/memory; artefacts (hands, text); bias and misuse concerns.
Applications	Compression, denoising, anomaly detection, pre-training.	Data generation, anomaly detection, latent interpolation, molecule design.	Data augmentation, face/art generation, representation learning.	Text-to-image, inpainting, super-resolution, design and media creation.
Reconstructed / generated output	Close copy of the input; slight loss of fine detail.	Smoother and blurrier reconstruction; new samples plausible but soft.	Sharp new images, but distortions or repeated patterns possible.	Detailed, photorealistic, prompt-matched images; occasional anatomy/text errors.

3. Task 2 – Practical Implementation and Output Analysis

Setup: Google Colab (T4 GPU), PyTorch. AE and VAE (latent size 16) and DCGAN were run on MNIST for 10 epochs; Stable Diffusion v1.5 was run via the Hugging Face *diffusers* library (30 steps, guidance 7.5). Notebook: Assignment02_GenAI_Colab.ipynb.

Activity A – Image reconstruction (CO3)



Reconstruction error (MSE on test images): AE = [0.011497630737721920], VAE = [0.015579163096845150].

- **Autoencoder:** reconstruction stays close to the input with sharper strokes; only fine detail is slightly lost. Decoding random latent vectors gives meaningless images, so it cannot generate.
- **VAE:** reconstruction is smoother and blurrier, with softened edges, because the KL term forces a regular latent space. Decoding random latent vectors gives plausible new digits.

Activity B – Image generation (CO4)




```
prompts = ['an astronaut riding a horse on the moon, digital art',
           'a photo of a golden retriever puppy playing in a park, high detail',
           'a handwritten digit 7 in black ink on white paper',
           'close-up of a person\'s hands typing on a keyboard']   # last one tests known weak spots (hands)
```

Prompt used: ["an astronaut riding a horse on the moon, digital art"]

- **DCGAN:** generates sharp digit-like images in one pass, but resolution is small, quality varies between samples, and some digits are distorted or malformed. It cannot be controlled by text.
- **Stable Diffusion:** produces detailed, high-resolution, prompt-matched images with realistic lighting and texture. Limitations: slower sampling (30 steps), GPU needed, and errors in hands, text and object counts.

Note: *observations above must be checked against our own screenshots and edited if our run differed.*

4. Task 3 – Group Reflection and AI Usage

- **AI tools used:** [ChatGPT] for explaining concepts, drafting the table and generating Colab code.
- **Important prompts:** (1) “Compare AE, VAE, DCGAN and Stable Diffusion on principle, architecture, advantages, limitations, applications and output differences.” (2) “Write a short PyTorch Colab notebook training AE and VAE on MNIST and plotting original vs reconstructions.” (3) “Write a DCGAN for MNIST and a Stable Diffusion pipeline with diffusers.” (4) “Explain why VAE outputs are blurrier than AE outputs.”
- **Verification:** checked AI statements against the original papers [1]–[5]; ran every notebook ourselves and compared our screenshots with the claims; edited statements that did not match our outputs.

- **Key learning:** reconstruction models (AE, VAE) learn compact representations, but only the VAE's regularised latent space supports sampling; GANs trade stability for sharpness; diffusion models give the best quality at higher compute cost.

Contributions:

- **Member 1 –** [Nikheel C Khadakabhavi]: Compared AE and VAE, ran Activity A and captured screenshots. Explained the KL term and reparameterisation in the presentation.
- **Member 2 –** [Gunjan Balapure]: Studied DCGAN, ran the DCGAN notebook and wrote the generation analysis. Prepared slides 3–4.Studied Stable Diffusion, ran the pretrained pipeline, tested prompts, and compiled the AI-usage log and final report.

References

[1] Kingma & Welling, Auto-Encoding Variational Bayes, 2014.
[2] Goodfellow et al., Generative Adversarial Nets, 2014.
[3] Radford et al., Unsupervised Representation Learning with DCGANs, ICLR 2016.
[4] Ho et al., Denoising Diffusion Probabilistic Models, 2020.
[5] Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, CVPR 2022.